



DEPARTMENT OF INFORMATION & COMMUNICATION TECHNOLOGY
SESSION II 2022/2023

DFP40203
PYTHON PROGRAMMING

CONTINUOUS ASSESSMENT:
PROBLEM BASED TASK 1

NAME	MATRIX NUMBER
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CLASS : DDT4A

LECTURER :
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a) Introduction :

Our problem-based task is to create a program that allows admins to manage books in the library.

The program should allow admin to add new books to the library, search for books by title or author, display all books in the library, and delete books from the library. The program should store the library data in a text file for purpose admin to manage it.

b) Coding :

```
TESTPBT.py X
1  # -*- coding: utf-8 -*-
2  """
3  TITLE : PROBLEM BASED TASK 1
4  NAME : 1) MUHAMMAD REDZUAN BIN PARSIKUN    MATRIX.NO : 1)03DDT21F1005
5          2) CHAIRIS A/L PUM                2)03DDT21F1088
6  CLASS: DDT4A
7
8  """
9  import datetime
10
11 #create the class of book
12 class Book:
13
14     # called every time an object is created from a class
15     # def __init__ as a method
16     def __init__(self, title, author, isbn, year_published):
17         self.title = title
18         self.author = author
19         self.isbn = isbn
20         self.year_published = year_published
21
22     #Refer to this coding when admin want to add,search,display and delete books
23     def __str__(self):
24         return f"Title: {self.title}\n Author: {self.author}\n ISBN: {self.isbn}\n Year Published: {self.year_published}\n"
25
26 #create class of libraray
27 class Library:
28     # def __init__ as a method
29     def __init__(self):
30         self.books = []
31
32     # def add_book as a method
33     def add_book(self, book):
34         self.books.append(book)
35
36     # def delete_book as a method
37     def delete_book(self, book):
```

```

38         self.books.remove(book)
39
40     # def search_book as a method
41     def search_book_by_title(self, title):
42         matching_books = []
43         for book in self.books:
44             if book.title.lower() == title.lower():
45                 matching_books.append(book)
46         return matching_books
47
48     # def display_all_book as a method
49     def display_all_books(self):
50         if len(self.books) == 0:
51             print("There are no books in the library.")
52         else:
53             for book in self.books:
54                 print(book)
55
56
57     #to read what admin create the file and directory handling
58     def read_library_file(file_path):
59         library = Library()
60         try:
61             with open(file_path, 'r') as file:
62                 for line in file:
63                     book_data = line.strip().split(',')
64                     book = Book(book_data[0], book_data[1], book_data[2], int(book_data[3]))
65                     library.add_book(book)
66         except FileNotFoundError:
67             pass
68         return library
69
70     #to create the file and directory handling
71     def write_library_file(file_path, library):
72         with open(file_path, 'w') as file:
73             for book in library.books:
74                 file.write(f"{book.title},{book.author},{book.isbn},{book.year_published}\n")

```

```

75
76
77
78     def add_books():
79
80         while True:
81             USER = input("Enter the book (yes) or quit?      :")
82
83             if USER == 'quit':
84                 break
85
86             elif USER == 'yes':
87                 # ask user for file name
88                 library_file_path = "Library.txt"
89
90                 library = read_library_file(library_file_path)
91                 title = input("Enter the title of the book      :")
92                 author = input("Enter the author of the book    :")
93                 isbn = input("Enter the ISBN of the book        :")
94                 year_published = int(input("Enter the year the book was published:"))
95                 book = Book(title, author, isbn, year_published)
96                 library.add_book(book)
97                 write_library_file(library_file_path, library)
98
99                 print("Book added to library.")
100                 print("++-----++")
101                 continue
102
103     def search_books():
104
105         # search for a book by title
106         title = input("Enter the title of the book to search for: ")
107         matching_books = library.search_book_by_title(title)
108
109         if len(matching_books) == 0:
110             print("\nNo matching books found.")
111

```

```

112         else:
113             print("\nMatching books:")
114             for i, book in enumerate(matching_books):
115                 print(f"{i+1}. {book}")
116
117
118 def menu():
119
120
121     while True:
122
123         print("++-----++")
124         print("+ 1 : Add Books          +")
125         print("+ 2 : Search Books         +")
126         print("+ 3 : Display                +")
127         print("+ 4 : Delete                 +")
128         print("+ 5 : Quit                   +")
129         print("++-----++")
130
131         num = int(input("Insert The Number : "))
132         if num == 1:
133             add_books()
134
135         elif num == 2:
136             search_books()
137         elif num == 3:
138             library.display_all_books()
139         elif num == 4:
140             title = input("Enter title of book to delete: ")
141             book_list = library.search_book_by_title(title)
142             if not book_list:
143                 print("Book not found")
144             else:
145                 for i, book in enumerate(book_list):
146                     print(i+1, book)
147                 book_num = int(input("Enter book number to delete: "))
148                 book = book_list[book_num-1]
149
150                 library.delete_book(book)
151                 write_library_file("Library.txt", library)
152                 print("Book deleted")
153         elif num == 5:
154             print("Thank you !!!")
155             break
156         else:
157             print("Wrong Input")
158             break
159
160 #display output
161
162 print("++-----++")
163 print("+ WELCOME TO PERPUSTAKAAN DESA +")
164 print("++-----++")
165
166 now = datetime.datetime.now()
167 print("Current date : ", now.strftime("%Y-%m-%d"))
168 print("Current time: ", now.strftime("%H:%M:%S"))
169 print("Today is : ", now.strftime("%A"))
170 print()
171
172 #declare object as file
173 library_file_path = "Library.txt"
174
175 #declare the object to read the file
176 library = read_library_file(library_file_path)
177
178 print("++-----++")
179 print("+ Admin Platform          +")
180 print("++-----++")
181 admin_Name = str(input("Admin's Name : "))
182 #make admin use this password if wrong it will display wrong output
183 while True:
184     admin_Password = str(input("Admin's Password : "))
185
186     if admin_Password == "DESAPerpustakaan23_":
187         menu()
188         break
189
190     elif admin_Password != "DESAPerpustakaan23_":
191         print("Wrong Password , Please Try Again")
192         continue

```

c) Output :

```
++-----++
+      WELCOME TO PERPUSTAKAAN DESA      +
++-----++
Current date : 2023-04-08
Current time: 16:26:16
Today is : Saturday

++-----++
+      Admin Platform      +
++-----++
Admin's Name : Muhammad Izzat
Admin's Password : DESAPerpustakaan33_
Wrong Password , Please Try Again
Admin's Password : DESAPerpustakaan23_
++-----++
+ 1 : Add Books      +
+ 2 : Search Books   +
+ 3 : Display        +
+ 4 : Delete         +
+ 5 : Quit           +
++-----++
Insert The Number : 1
Enter the book (yes) or quit?      :yes
Enter the title of the book        :Panglima Bersatu
Enter the author of the book       :Jamal Ainuddin
Enter the ISBN of the book         :334466
Enter the year the book was published:2012
Book added to library.

++-----++
Enter the book (yes) or quit?      :yes
Enter the title of the book        :Isu Remaja
Enter the author of the book       :Izatti Mazlan
Enter the ISBN of the book         :466782
Enter the year the book was published:2011
Book added to library.
++-----++
Enter the book (yes) or quit?      :quit
++-----++
+ 1 : Add Books      +
+ 2 : Search Books   +
+ 3 : Display        +
+ 4 : Delete         +
+ 5 : Quit           +
++-----++
```

```
Insert The Number : 2
Enter the title of the book to search for: Panglima Bersatu
```

```
Matching books:
1. Title: Panglima Bersatu
   Author: Jamal Ainuddin
   ISBN: 334466
   Year Published: 2012
```

```
++-----++
+ 1 : Add Books          +
+ 2 : Search Books       +
+ 3 : Display            +
+ 4 : Delete             +
+ 5 : Quit               +
++-----++
```

```
Insert The Number : 3
Title: Panglima Bersatu
Author: Jamal Ainuddin
```

```
ISBN: 334466
Year Published: 2012
```

```
Title: Isu Remaja
Author: Izatti Mazlan
ISBN: 466782
Year Published: 2011
```

```
++-----++
+ 1 : Add Books          +
+ 2 : Search Books       +
+ 3 : Display            +
+ 4 : Delete             +
+ 5 : Quit               +
++-----++
```

```
Insert The Number : 4
Enter title of book to delete: Isu Remaja
```

```
1 Title: Isu Remaja
   Author: Izatti Mazlan
   ISBN: 466782
   Year Published: 2011
```

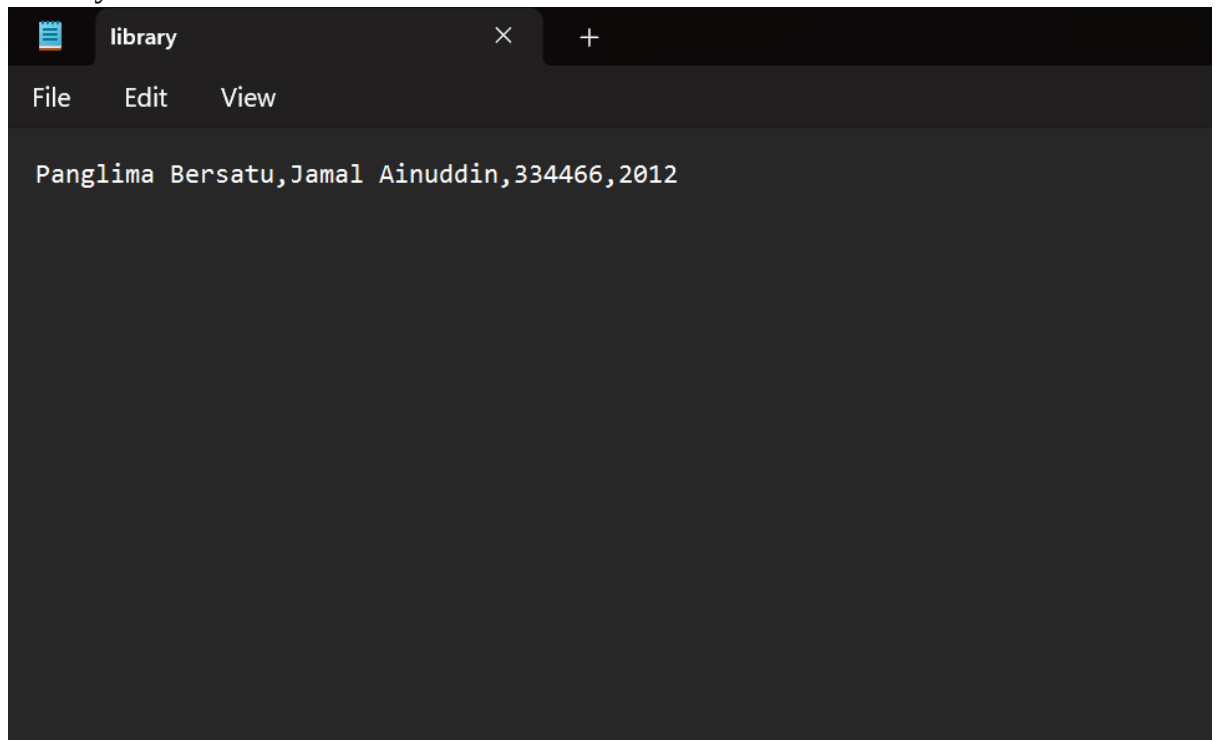
```
Enter book number to delete: 1
Book deleted
```

```
++-----++
+ 1 : Add Books          +
+ 2 : Search Books       +
+ 3 : Display            +
+ 4 : Delete             +
```

```
+ 5 : Quit              +
++-----++
```

```
Insert The Number : 5
Thank you !!!
```

Library.txt



d) Conclusion:

In conclusion, in this assignment, we can learn and understand file and directory handling with function and Object-Oriented Programming (OOP) to develop project which is library management.

With file directory handling, the admin is able to use the system clearly because it can show the contents of the text files that the admin stores. In addition, admins are also able to search for book titles that they have saved. In addition, the admin can also show all the books in the file-directory. Finally, it can also delete the file-directory and make the data disappear.